BEST KNOWN SQUARE PACKINGS

The square packing problem asks for the side *s*(*n*) of the smallest square that can hold *n* unit squares, where the squares are free to rotate but cannot overlap.

- O proved optimal (77)
- = exact value known (292)
- ≈ only known numerically (32)
- R rigid (22)

- ★ recent result, since August, 2026 (297)
- 90° 45° 🟪 🟦 colors indicate distinct tilt angles
- 4 3 2 1 0 shade indicates number of full-side contacts
- deg is the algebraic degree of that side length

Best packings due to Chaoweeraprasit, Daniel, Rehwaldt, Couzo, de Winter, Schadt, Cantrell, Ellsworth, Hajba, DeVincentis, Morandi, Bidwell, Friedman, Hämäläinen, Stenlund, Göbel, Trump, Wainwright, Chang, hmbelvedere.

Diagram by Joshua Levy

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The Squares Project

github.com/jlevy/squares

1×1 GRID
1 O = R s(1) = 1
2×2 GRID
2 O = s(2) = 2
3 O = s(3) = 2
4 O = R s(4) = 2
5 O = R s(5) = 2.70711 deg 2
3×3 GRID
6 O = s(6) = 3
7 ★ O = s(7) = 3
8 ★ O = s(8) = 3
9 O = R s(9) = 3
10 O = s(10) = 3.70711 deg 2
11 ★ O = R s(11) = 3.87708 deg 8
4×4 GRID
12 ★ = s(12) ≤ 4 s(12) ≥ 3.97150
13 O = s(13) = 4
14 ★ O = s(14) = 4
15 ★ O = s(15) = 4
16 O = R s(16) = 4
17 ★ = s(17) ≤ 4.67554 deg 18 s(17) ≥ 4.66044
18 ★ = s(18) ≤ 4.82288 deg 2 s(18) ≥ 4.70500
19 ★ = s(19) ≤ 4.88562 deg 2 s(19) ≥ 4.82500
5×5 GRID
20 ★ = s(20) ≤ 5 s(20) ≥ 4.90500
21 ★ O = s(21) = 5
22 O = s(22) = 5
23 ★ O = s(23) = 5
24 ★ O = s(24) = 5
25 O = R s(25) = 5
26 ★ = s(26) ≤ 5.62133 deg 2 s(26) ≥ 5.54500
27 ★ = s(27) ≤ 5.70711 deg 2 s(27) ≥ 5.64350
28 ★ = R s(28) ≤ 5.82445 deg 6 s(28) ≥ 5.73500
29 ★ ≈ s(29) ≤ 5.93384 s(29) ≥ 5.81000
6×6 GRID
30 ★ = s(30) ≤ 6 s(30) ≥ 5.88350
31 ★ = s(31) ≤ 6 s(31) ≥ 5.95250
32 ★ O = s(32) = 6
33 O = s(33) = 6
34 ★ O = s(34) = 6
35 ★ O = s(35) = 6
36 O = R s(36) = 6
37 ★ = s(37) ≤ 6.59862 deg 8 s(37) ≥ 6.44000
38 ★ = s(38) ≤ 6.70711 deg 2 s(38) ≥ 6.54500
39 ★ = s(39) ≤ 6.81073 deg 5 s(39) ≥ 6.65000
40 ★ = R s(40) ≤ 6.82843 deg 2 s(40) ≥ 6.70000
41 ★ = s(41) ≤ 6.92670 deg 42 s(41) ≥ 6.77500
7×7 GRID
42 ★ = s(42) ≤ 7 s(42) ≥ 6.84750
43 ★ = s(43) ≤ 7 s(43) ≥ 6.90750
44 ★ = s(44) ≤ 7 s(44) ≥ 6.97250
45 ★ O = s(45) = 7
46 O = s(46) = 7
47 ★ O = s(47) = 7
48 ★ O = s(48) = 7
49 O = R s(49) = 7
50 ★ = s(50) ≤ 7.57143 s(50) ≥ 7.40000
51 ★ = s(51) ≤ 7.70080 deg 12 s(51) ≥ 7.47000
52 ★ = s(52) ≤ 7.70711 deg 2 s(52) ≥ 7.55000
53 ★ = s(53) ≤ 7.82288 deg 2 s(53) ≥ 7.62750
54 ★ = s(54) ≤ 7.84667 deg 4 s(54) ≥ 7.68500
55 ★ ≈ s(55) ≤ 7.94578 s(55) ≥ 7.72800
8×8 GRID
56 ★ = s(56) ≤ 8 s(56) ≥ 7.80250
57 ★ = s(57) ≤ 8 s(57) ≥ 7.87250
58 ★ = s(58) ≤ 8 s(58) ≥ 7.93500
59 ★ O = s(59) = 8
60 ★ O = s(60) = 8
61 ★ O = s(61) = 8
62 ★ O = s(62) = 8
63 ★ O = s(63) = 8
64 O = R s(64) = 8
65 ★ = s(65) ≤ 8.53554 deg 2 s(65) ≥ 8.35000
66 ★ = s(66) ≤ 8.65686 deg 2 s(66) ≥ 8.43000
67 ★ = s(67) ≤ 8.70711 deg 2 s(67) ≥ 8.48000
68 ★ = s(68) ≤ 8.79880 s(68) ≥ 8.51000
69 ★ = s(69) ≤ 8.82720 deg 38 s(69) ≥ 8.62000
70 ★ = s(70) ≤ 8.88167 deg 4 s(70) ≥ 8.65750
71 ★ ≈ s(71) ≤ 8.94408 s(71) ≥ 8.72100
9×9 GRID
72 ★ = s(72) ≤ 9 s(72) ≥ 8.76000
73 ★ = s(73) ≤ 9 s(73) ≥ 8.81300
74 ★ = s(74) ≤ 9 s(74) ≥ 8.86750
75 ★ = s(75) ≤ 9 s(75) ≥ 8.94000
76 ★ = s(76) ≤ 9 s(76) ≥ 8.96500
77 ★ O = s(77) = 9
78 ★ O = s(78) = 9
79 ★ O = s(79) = 9
80 ★ O = s(80) = 9
81 O = R s(81) = 9
82 ★ = s(82) ≤ 9.53554 deg 2 s(82) ≥ 9.32000
83 ★ = s(83) ≤ 9.63476 deg 672 s(83) ≥ 9.37000
84 ★ = s(84) ≤ 9.70711 deg 2 s(84) ≥ 9.41100
85 ★ = s(85) ≤ 9.74265 deg 2 s(85) ≥ 9.46000
86 ★ = s(86) ≤ 9.82288 deg 2 s(86) ≥ 9.50300
87 ★ = s(87) ≤ 9.83882 deg 41 s(87) ≥ 9.58000
88 ★ = s(88) ≤ 9.88246 s(88) ≥ 9.62000
89 ★ = s(89) ≤ 9.94975 deg 2 s(89) ≥ 9.65000
10×10 GRID
90 ★ = s(90) ≤ 10 s(90) ≥ 9.73000
91 ★ = s(91) ≤ 10 s(91) ≥ 9.76250
92 ★ = s(92) ≤ 10 s(92) ≥ 9.77000
93 ★ = s(93) ≤ 10 s(93) ≥ 9.88000
94 ★ = s(94) ≤ 10 s(94) ≥ 9.95000
95 ★ = s(95) ≤ 10 s(95) ≥ 9.96500
96 ★ = s(96) ≤ 10 s(96) ≥ 9.97000
97 ★ O = s(97) = 10
98 ★ O = s(98) = 10
99 ★ O = s(99) = 10
100 O = R s(100) = 10
101 ★ = s(101) ≤ 10.53554 deg 2 s(101) ≥ 10.28000
102 ★ ≈ s(102) ≤ 10.60718 s(102) ≥ 10.28000
103 ★ ≈ s(103) ≤ 10.70352 s(103) ≥ 10.28000
104 ★ = s(104) ≤ 10.70711 deg 2 s(104) ≥ 10.28000
105 ★ = s(105) ≤ 10.80608 s(105) ≥ 10.28000
106 ★ ≈ s(106) ≤ 10.82291 s(106) ≥ 10.31070
107 ★ = s(107) ≤ 10.84667 deg 4 s(107) ≥ 10.36154
108 ★ = s(108) ≤ 10.90995 s(108) ≥ 10.41211
109 ★ = s(109) ≤ 10.94975 deg 2 s(109) ≥ 10.46242
110 ★ ≈ s(110) ≤ 10.99679 s(110) ≥ 10.51249
11×11 GRID
111 ★ = s(111) ≤ 11 s(111) ≥ 10.56230
112 ★ = s(112) ≤ 11 s(112) ≥ 10.61187
113 ★ = s(113) ≤ 11 s(113) ≥ 10.66120
114 ★ = s(114) ≤ 11 s(114) ≥ 10.71028
115 ★ = s(115) ≤ 11 s(115) ≥ 10.75914
116 ★ = s(116) ≤ 11 s(116) ≥ 10.80776
117 ★ = s(117) ≤ 11 s(117) ≥ 10.85615
118 ★ O = s(118) = 11
119 ★ O = s(119) = 11
120 ★ O = s(120) = 11
121 O = R s(121) = 11
122 ★ = s(122) ≤ 11.53554 deg 2 s(122) ≥ 11.04751
123 ★ = s(123) ≤ 11.60091 s(123) ≥ 11.09481
124 ★ = s(124) ≤ 11.65686 deg 2 s(124) ≥ 11.14189
125 ★ = s(125) ≤ 11.70711 deg 2 s(125) ≥ 11.18877
126 ★ = s(126) ≤ 11.77331 s(126) ≥ 11.23545
127 ★ = s(127) ≤ 11.82288 deg 2 s(127) ≥ 11.28192
128 ★ = s(128) ≤ 11.8251 deg 40 s(128) ≥ 11.32820
129 ★ = s(129) ≤ 11.87938 s(129) ≥ 11.37428
130 ★ = s(130) ≤ 11.90449 s(130) ≥ 11.42016
131 ★ ≈ s(131) ≤ 11.95492 s(131) ≥ 11.46585
132 ★ ≈ s(132) ≤ 11.99133 s(132) ≥ 11.51135
12×12 GRID
133 ★ = s(133) ≤ 12 s(133) ≥ 11.55667
134 ★ = s(134) ≤ 12 s(134) ≥ 11.60180
135 ★ = s(135) ≤ 12 s(135) ≥ 11.64674
136 ★ = s(136) ≤ 12 s(136) ≥ 11.69151
137 ★ = s(137) ≤ 12 s(137) ≥ 11.73610
138 ★ = s(138) ≤ 12 s(138) ≥ 11.78051
139 ★ = s(139) ≤ 12 s(139) ≥ 11.82475
140 ★ = s(140) ≤ 12 s(140) ≥ 11.86881
141 ★ O = s(141) = 12
142 ★ O = s(142) = 12
143 ★ O = s(143) = 12
144 O = R s(144) = 12
145 ★ = s(145) ≤ 12.53554 deg 2 s(145) ≥ 12.04339
146 ★ = s(146) ≤ 12.6010 deg 16 s(146) ≥ 12.08663
147 ★ = s(147) ≤ 12.65686 deg 2 s(147) ≥ 12.12970
148 ★ = s(148) ≤ 12.65686 deg 2 s(148) ≥ 12.17261
149 ★ = s(149) ≤ 12.70711 deg 2 s(149) ≥ 12.21537
150 ★ = s(150) ≤ 12.77818 deg 2 s(150) ≥ 12.25797
151 ★ = s(151) ≤ 12.82288 deg 2 s(151) ≥ 12.30042
152 ★ ≈ s(152) ≤ 12.83072 s(152) ≥ 12.34271
153 ★ = s(153) ≤ 12.87968 s(153) ≥ 12.38486
154 ★ = s(154) ≤ 12.92657 s(154) ≥ 12.42686
155 ★ = s(155) ≤ 12.95251 s(155) ≥ 12.46870
156 ★ ≈ s(156) ≤ 12.98209 s(156) ≥ 12.51041
13×13 GRID
157 ★ = s(157) ≤ 13 s(157) ≥ 12.55197
158 ★ = s(158) ≤ 13 s(158) ≥ 12.59338
159 ★ = s(159) ≤ 13 s(159) ≥ 12.63466
160 ★ = s(160) ≤ 13 s(160) ≥ 12.67579
161 ★ = s(161) ≤ 13 s(161) ≥ 12.71679
162 ★ = s(162) ≤ 13 s(162) ≥ 12.75765
163 ★ = s(163) ≤ 13 s(163) ≥ 12.79837
164 ★ = s(164) ≤ 13 s(164) ≥ 12.83896
165 ★ = s(165) ≤ 13 s(165) ≥ 12.87941
166 ★ O = s(166) = 13
167 ★ O = s(167) = 13
168 ★ O = s(168) = 13
169 O = R s(169) = 13
170 ★ = s(170) ≤ 13.53554 deg 2 s(170) ≥ 13.03993
171 ★ = s(171) ≤ 13.57143 s(171) ≥ 13.07974
172 ★ ≈ s(172) ≤ 13.61899 s(172) ≥ 13.11942
173 ★ = s(173) ≤ 13.65686 deg 2 s(173) ≥ 13.15898
174 ★ = s(174) ≤ 13.70711 deg 2 s(174) ≥ 13.19842
175 ★ = s(175) ≤ 13.77818 deg 2 s(175) ≥ 13.23773
176 ★ = s(176) ≤ 13.82288 deg 2 s(176) ≥ 13.27693
177 ★ ≈ s(177) ≤ 13.82298 s(177) ≥ 13.31600
178 ★ = s(178) ≤ 13.84667 deg 4 s(178) ≥ 13.35496
179 ★ = s(179) ≤ 13.88380 s(179) ≥ 13.39379
180 ★ = s(180) ≤ 13.91766 s(180) ≥ 13.43251
181 ★ ≈ s(181) ≤ 13.95375 s(181) ≥ 13.47112
182 ★ ≈ s(182) ≤ 13.97410 s(182) ≥ 13.50961
14×14 GRID
183 ★ = s(183) ≤ 14 s(183) ≥ 13.54798
184 ★ = s(184) ≤ 14 s(184) ≥ 13.58625
185 ★ = s(185) ≤ 14 s(185) ≥ 13.62440
186 ★ = s(186) ≤ 14 s(186) ≥ 13.66244
187 ★ = s(187) ≤ 14 s(187) ≥ 13.70037
188 ★ = s(188) ≤ 14 s(188) ≥ 13.73820
189 ★ = s(189) ≤ 14 s(189) ≥ 13.77591
190 ★ = s(190) ≤ 14 s(190) ≥ 13.81352
191 ★ = s(191) ≤ 14 s(191) ≥ 13.85102
192 ★ = s(192) ≤ 14 s(192) ≥ 13.88842
193 ★ O = s(193) = 14
194 ★ O = s(194) = 14
195 ★ O = s(195) = 14
196 O = R s(196) = 14
197 ★ = s(197) ≤ 14.53554 deg 2 s(197) ≥ 14.03698
198 ★ = s(198) ≤ 14.57143 s(198) ≥ 14.07387
199 ★ = s(199) ≤ 14.61758 s(199) ≥ 14.11065
200 ★ = s(200) ≤ 14.65686 deg 2 s(200) ≥ 14.14734
201 ★ = s(201) ≤ 14.70711 deg 2 s(201) ≥ 14.18393
202 ★ = s(202) ≤ 14.72793 deg 2 s(202) ≥ 14.22042
203 ★ = s(203) ≤ 14.77818 deg 2 s(203) ≥ 14.25681
204 ★ = s(204) ≤ 14.82288 deg 2 s(204) ≥ 14.29311
205 ★ = s(205) ≤ 14.8245 deg 40 s(205) ≥ 14.32931
206 ★ ≈ s(206) ≤ 14.86016 s(206) ≥ 14.36542
207 ★ = s(207) ≤ 14.88800 s(207) ≥ 14.40143
208 ★ = s(208) ≤ 14.92452 s(208) ≥ 14.43735
209 ★ = s(209) ≤ 14.94962 s(209) ≥ 14.47318
210 ★ ≈ s(210) ≤ 14.97301 s(210) ≥ 14.50892
211 ★ ≈ s(211) ≤ 14.99797 s(211) ≥ 14.54457
15×15 GRID
212 ★ = s(212) ≤ 15 s(212) ≥ 14.58012
213 ★ = s(213) ≤ 15 s(213) ≥ 14.61559
214 ★ = s(214) ≤ 15 s(214) ≥ 14.65097
215 ★ = s(215) ≤ 15 s(215) ≥ 14.68626
216 ★ = s(216) ≤ 15 s(216) ≥ 14.72146
217 ★ = s(217) ≤ 15 s(217) ≥ 14.75657
218 ★ = s(218) ≤ 15 s(218) ≥ 14.79160
219 ★ = s(219) ≤ 15 s(219) ≥ 14.82654
220 ★ = s(220) ≤ 15 s(220) ≥ 14.86140
221 ★ = s(221) ≤ 15 s(221) ≥ 14.89618
222 ★ O = s(222) = 15
223 ★ O = s(223) = 15
224 ★ O = s(224) = 15
225 O = R s(225) = 15
226 ★ = s(226) ≤ 15.53554 deg 2 s(226) ≥ 15.03444
227 ★ = s(227) ≤ 15.57107 deg 2 s(227) ≥ 15.06880
228 ★ ≈ s(228) ≤ 15.60461 s(228) ≥ 15.10308
229 ★ = s(229) ≤ 15.65686 deg 2 s(229) ≥ 15.13728
230 ★ = s(230) ≤ 15.68293 s(230) ≥ 15.17140
231 ★ = s(231) ≤ 15.70711 deg 2 s(231) ≥ 15.20544
232 ★ = s(232) ≤ 15.77818 deg 2 s(232) ≥ 15.23940
233 ★ = s(233) ≤ 15.77818 deg 2 s(233) ≥ 15.27328
234 ★ = s(234) ≤ 15.82288 deg 2 s(234) ≥ 15.30709
235 ★ = s(235) ≤ 15.8267 deg 83 s(235) ≥ 15.34082
236 ★ = s(236) ≤ 15.86781 s(236) ≥ 15.37447
237 ★ = s(237) ≤ 15.90368 s(237) ≥ 15.40805
238 ★ = s(238) ≤ 15.92615 s(238) ≥ 15.44155
239 ★ = s(239) ≤ 15.94932 s(239) ≥ 15.47497
240 ★ ≈ s(240) ≤ 15.96969 s(240) ≥ 15.50833
241 ★ ≈ s(241) ≤ 15.98814 s(241) ≥ 15.54160
16×16 GRID
242 ★ = s(242) ≤ 16 s(242) ≥ 15.57481
243 ★ = s(243) ≤ 16 s(243) ≥ 15.60794
244 ★ = s(244) ≤ 16 s(244) ≥ 15.64100
245 ★ = s(245) ≤ 16 s(245) ≥ 15.67399
246 ★ = s(246) ≤ 16 s(246) ≥ 15.70690
247 ★ = s(247) ≤ 16 s(247) ≥ 15.73975
248 ★ = s(248) ≤ 16 s(248) ≥ 15.77252
249 ★ = s(249) ≤ 16 s(249) ≥ 15.80522
250 ★ = s(250) ≤ 16 s(250) ≥ 15.83786
251 ★ = s(251) ≤ 16 s(251) ≥ 15.87042
252 ★ = s(252) ≤ 16 s(252) ≥ 15.90292
253 ★ O = s(253) = 16
254 ★ O = s(254) = 16
255 ★ O = s(255) = 16
256 O = R s(256) = 16
257 ★ = s(257) ≤ 16.53554 deg 2 s(257) ≥ 16.03222
258 ★ = s(258) ≤ 16.56345 s(258) ≥ 16.06438
259 ★ ≈ s(259) ≤ 16.60257 s(259) ≥ 16.09647
260 ★ = s(260) ≤ 16.65686 deg 2 s(260) ≥ 16.12849
261 ★ = s(261) ≤ 16.68293 s(261) ≥ 16.16045
262 ★ = s(262) ≤ 16.70711 deg 2 s(262) ≥ 16.19235
263 ★ = s(263) ≤ 16.74042 s(263) ≥ 16.22418
264 ★ = s(264) ≤ 16.77818 deg 2 s(264) ≥ 16.25595
265 ★ = s(265) ≤ 16.77818 deg 2 s(265) ≥ 16.28765
266 ★ = s(266) ≤ 16.82303 s(266) ≥ 16.31929
267 ★ = s(267) ≤ 16.84667 deg 4 s(267) ≥ 16.35086
268 ★ ≈ s(268) ≤ 16.87882 s(268) ≥ 16.38238
269 ★ ≈ s(269) ≤ 16.90597 s(269) ≥ 16.41383
270 ★ = s(270) ≤ 16.93781 s(270) ≥ 16.44521
271 ★ ≈ s(271) ≤ 16.95083 s(271) ≥ 16.47654
272 ★ = s(272) ≤ 16.96812 s(272) ≥ 16.50781
273 ★ ≈ s(273) ≤ 16.98393 s(273) ≥ 16.53901
17×17 GRID
274 ★ = s(274) ≤ 17 s(274) ≥ 16.57015
275 ★ = s(275) ≤ 17 s(275) ≥ 16.60124
276 ★ = s(276) ≤ 17 s(276) ≥ 16.63226
277 ★ = s(277) ≤ 17 s(277) ≥ 16.66322
278 ★ = s(278) ≤ 17 s(278) ≥ 16.69413
279 ★ = s(279) ≤ 17 s(279) ≥ 16.72498
280 ★ = s(280) ≤ 17 s(280) ≥ 16.75576
281 ★ = s(281) ≤ 17 s(281) ≥ 16.78649
282 ★ = s(282) ≤ 17 s(282) ≥ 16.81716
283 ★ = s(283) ≤ 17 s(283) ≥ 16.84778
284 ★ = s(284) ≤ 17 s(284) ≥ 16.87833
285 ★ = s(285) ≤ 17 s(285) ≥ 16.90883
286 ★ O = s(286) = 17
287 ★ O = s(287) = 17
288 ★ O = s(288) = 17
289 O = R s(289) = 17
290 ★ = s(290) ≤ 17.53554 deg 2 s(290) ≥ 17.03027
291 ★ = s(291) ≤ 17.53554 deg 2 s(291) ≥ 17.06049
292 ★ = s(292) ≤ 17.59725 s(292) ≥ 17.09066
293 ★ = s(293) ≤ 17.63415 s(293) ≥ 17.12077
294 ★ = s(294) ≤ 17.65686 deg 2 s(294) ≥ 17.15082
295 ★ = s(295) ≤ 17.70711 deg 2 s(295) ≥ 17.18082
296 ★ = s(296) ≤ 17.70711 deg 2 s(296) ≥ 17.21077
297 ★ ≈ s(297) ≤ 17.74042 s(297) ≥ 17.24066
298 ★ = s(298) ≤ 17.77818 deg 2 s(298) ≥ 17.27050
299 ★ = s(299) ≤ 17.82288 deg 2 s(299) ≥ 17.30029
300 ★ = s(300) ≤ 17.8242 deg 40 s(300) ≥ 17.33003
301 ★ ≈ s(301) ≤ 17.84667 s(301) ≥ 17.35971
302 ★ = s(302) ≤ 17.88131 s(302) ≥ 17.38934
303 ★ = s(303) ≤ 17.92032 s(303) ≥ 17.41892
304 ★ ≈ s(304) ≤ 17.93466 s(304) ≥ 17.44845
305 ★ ≈ s(305) ≤ 17.95296 s(305) ≥ 17.47792
306 ★ ≈ s(306) ≤ 17.96344 s(306) ≥ 17.50735
307 ★ ≈ s(307) ≤ 17.98104 s(307) ≥ 17.53672
18×18 GRID
308 ★ = s(308) ≤ 18 s(308) ≥ 17.56604
309 ★ = s(309) ≤ 18 s(309) ≥ 17.59532
310 ★ = s(310) ≤ 18 s(310) ≥ 17.62454
311 ★ = s(311) ≤ 18 s(311) ≥ 17.65371
312 ★ = s(312) ≤ 18 s(312) ≥ 17.68284
313 ★ = s(313) ≤ 18 s(313) ≥ 17.71191
314 ★ = s(314) ≤ 18 s(314) ≥ 17.74093
315 ★ = s(315) ≤ 18 s(315) ≥ 17.76991
316 ★ = s(316) ≤ 18 s(316) ≥ 17.79884
317 ★ = s(317) ≤ 18 s(317) ≥ 17.82772
318 ★ = s(318) ≤ 18 s(318) ≥ 17.85655
319 ★ = s(319) ≤ 18 s(319) ≥ 17.88533
320 ★ = s(320) ≤ 18 s(320) ≥ 17.91407
321 ★ O = s(321) = 18
322 ★ O = s(322) = 18
323 ★ O = s(323) = 18
324 O = R s(324) = 18